WASP-52b

R-1599

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_220911` · created 2026-08-02T02:07:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459833.9346 +/- 0.0068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.063
Transit depth (Rp/Rs)^2	4.66 +/- 2.74 [%]
Orbital Inclination (inc)	87.71 +/- 2.76
Airmass coefficient 1 (a1)	1.17 +/- 0.67
Airmass coefficient 2 (a2)	-0.19 +/- 0.3
Scatter in the residuals of the lightcurve fit is	49.41 %
Best Comparison Star	#1 - [541, 437]
Optimal Aperture	1.11
Optimal Annulus	5.5
Transit Duration (day)	0.084 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.063 vs 0.1646 (deviation 0.66σ)
Duration fit vs published	0.084 d vs 0.0758 d (deviation 0.47σ)
Mid-time O–C	12.0 min
Depth SNR	2.8
Residual scatter	49.41 %

Light curve

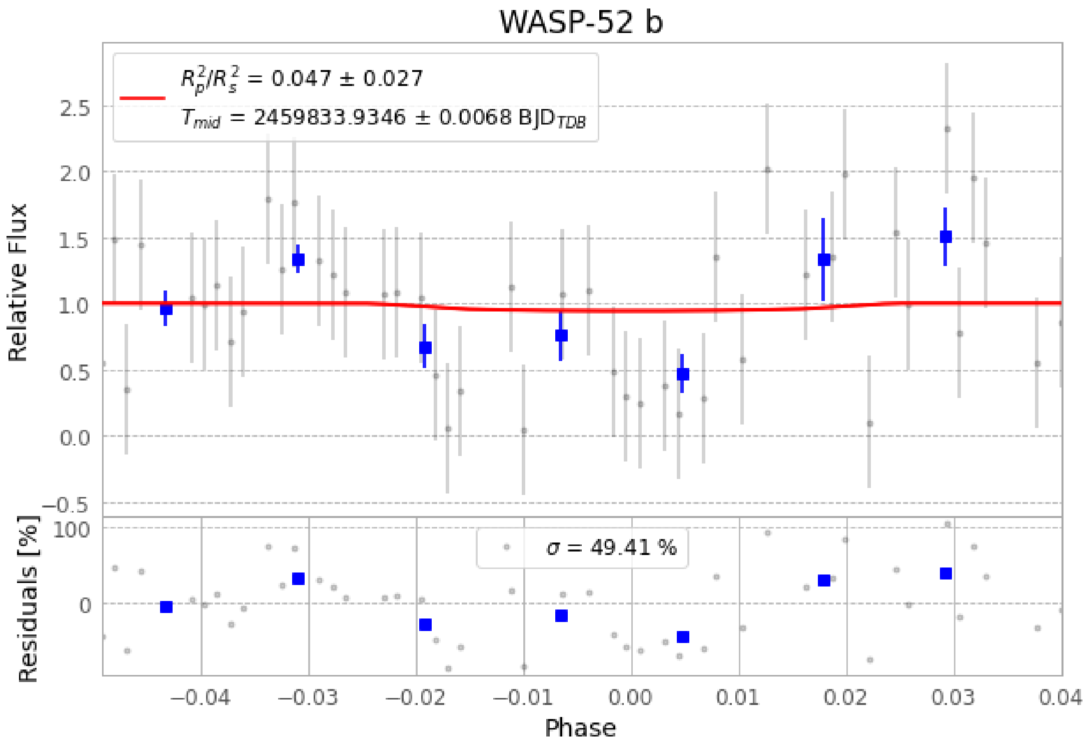

Figure 1 — FinalLightCurve_WASP-52 b_2022-09-11.png (SHA-256 e803dff79e80f0ce…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 334], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 eff6f12504d3ac3d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `WASP-52220911081213.FITS` → `WASP-52220911120013.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-220911.log (f300f4d1c4b0…), FinalLightCurve_WASP-52 b_2022-09-11.png (e803dff79e80…), FinalLightCurve_WASP-52 b_2022-09-11.csv (931a0df3053b…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)